PRODUCT SALES ANALYSIS

A COMPREHENSIVE SQL PROJECT

DATA TABLE DESCRIPTION

1. THIS DATASET PROVIDES COMPREHENSIVE INFORMATION ABOUT PRODUCT SALES PERFORMANCE.
2. IT INCLUDES PRODUCT DETAILS, PRICING, DISCOUNTS, SALES REVENUE, AND PROFITABILITY METRICS.
3. THE DATA HELPS IN IDENTIFYING HIGH-PERFORMING PRODUCTS, EVALUATING DISCOUNT IMPACT, AND ANALYZING OVERALL SALES TRENDS.
4. IT ENABLES COMPARATIVE ANALYSIS BETWEEN DIFFERENT PRODUCTS AND CATEGORIES TO IDENTIFY GROWTH OPPORTUNITIES AND AREAS THAT REQUIRE IMPROVEMENT.

GOALS AND FOCUS

PROJECT GOALS:

THE MAIN GOAL OF THIS PROJECT IS TO ANALYZE PRODUCT SALES PERFORMANCE AND IDENTIFY HIGH, MEDIUM, AND LOW-PERFORMING PRODUCTS. IT ALSO FOCUSES ON EVALUATING THE IMPACT OF DISCOUNTS AND PROFITABILITY TO SUPPORT BETTER BUSINESS DECISIONS.

KEY FOCUS AREA:

1. THE KEY FOCUS OF THIS DATASET IS TO ANALYZE PRODUCT SALES PERFORMANCE, REVENUE GENERATION, AND PROFITABILITY ACROSS DIFFERENT CATEGORIES.
2. IT AIMS TO IDENTIFY TOP-PERFORMING AND UNDERPERFORMING PRODUCTS WHILE EVALUATING THE IMPACT OF DISCOUNTS ON OVERALL BUSINESS GROWTH.

PROCESS CONDUCTED

- **DATA COLLECTION:** GATHER PRODUCT SALES DATA INCLUDING PRODUCT DETAILS, SALES, DISCOUNTS, AND PROFIT INFORMATION.
- **DATA CLEANING:** HANDLE MISSING VALUES, CORRECT DATA TYPES, AND REMOVE DUPLICATES TO ENSURE DATA ACCURACY.
- **DATA TRANSFORMATION:** CREATE CALCULATED COLUMNS AND MEASURES FOR REVENUE, PROFIT, AND PERFORMANCE ANALYSIS.
- **DATA ANALYSIS:** IDENTIFY HIGH, MEDIUM, AND LOW-PERFORMING PRODUCTS USING KEY METRICS.
- **VISUALIZATION & REPORTING:** BUILD DASHBOARDS TO PRESENT INSIGHTS CLEARLY FOR BUSINESS DECISION-MAKING.

1. HOW MANY PRODUCT CATEGORY ARE THERE ?

```
select
product_category
from productsb
group by product_category;
```

product_category
Furniture
Food
Clothing
Electronics

- This dataset have four product category that is Furniture, Food, Clothing, Electronics

2. HOW MANY QUANTITY ARE SOLD IN EACH PRODUCT CATEGORY ?

```
select
product_category,
count(quantity_sold)
from productsb
group by product_category;
```

	product_category	count(quantity_sold)
▶	Furniture	260
	Food	226
	Clothing	268
	Electronics	246

- The Product of Furniture is high sold of quantity compared to another product.

3. WHAT IS THE TOTAL SALES OF EACH PRODUCT ?

```sql
select
product_category,
sum(quantity_sold * unit_price) as Revenue
from productsb
group by product_category
order by Revenue desc;
```

product_category	Revenue
Clothing	19286733.929999996
Furniture	18330280.560000006
Electronics	17571040.570000004
Food	15141885.649999999

- The clothing product is number one sold in compared to each other product

4. WHICH PRODUCT HAS THE AVERAGE TOTAL SALES VALUE?

```
select product_category,
avg(quantity_sold*unit_price) as Total_sales
from productsb
group by product_category;
```

product_category	Total_sales
Furniture	70501.0790769231
Food	66999.49402654867
Clothing	71965.42511194029
Electronics	71426.99418699188

- Furniture product is the average of total sales Rs. 70501.079

5. WHICH CUSTOMER TYPE IS MOST PURCHASE BY THE CUSTOMERS?

```
select
product_category,
customer_type,
sum(quantity_sold)
from productsb
group by product_category,customer_type;
```

product_category	customer_type	sum(quantity_sold)
Furniture	Returning	3759
Food	Returning	2706
Clothing	New	3895
Electronics	New	3417
Food	New	2902
Furniture	New	2970
Clothing	Returning	3027
Electronics	Returning	2679

- The Returning customer type is most purchased by the customers

6. WHAT IS THE TOTAL REVENUE GENERATED BY EACH REGION?

```sql
select
Region,
sum(quantity_sold*unit_price) as Total_Revenue
from productsb
group by region;
```

Region	Total_Revenue
North	18208403.850000013
West	17761568.64999999
South	16282567.160000017
East	18077401.05000001

- North Region is highest total Revenue to compared with another regions

7. WHAT IS THE TOTAL SALES IN EACH SALES REP?

```
select
sales_rep,
sum(quantity_sold*unit_price) as Total_sales
from productsb
group by sales_rep
order by Total_sales desc;
```

sales_rep	Total_sales
David	16395583.950000012
Eve	15271749.440000003
Bob	13712628.180000016
Alice	13397730.089999998
Charlie	11552249.049999993

- The sales rep David is a number one total sales Rs. 16395583.95

8. WHICH SALES REP IS MAXIMUM USE IN RETAIL SALES CHANNEL?

```
select
sales_rep,
sales_channel,
max(quantity_sold*unit_price) as Total_sales
from productsb
where sales_channel="retail"
group by sales_rep
order by Total_sales desc;
```

	sales_rep	sales_channel	Total_sales
▶	Eve	Retail	233903.96000000002
	David	Retail	231204.74
	Alice	Retail	230817.84
	Bob	Retail	218146.27
	Charlie	Retail	213227.53999999998

- The Eve sales rep is maximum of use in retail sales channel.

9. WHICH PRODUCT ID IS HIGHEST TOTAL SALES?

```sql
select
ï»¿Product_ID,
sum(quantity_sold*unit_price) as Total_sales
from productsb
group by ï»¿Product_ID
order by Total_sales desc;
```

ï»¿Product_ID	Total_sales
1090	1819857.9099999997
1092	1738987.88
1099	1635859.89
1089	1445192.74
1054	1256881.4000000001
1058	1205704.4000000001
1062	1192222.3800000001
1061	1097459.1099999999
1008	1092001.7999999998
1075	1069940.3900000001
1063	1068608.51
1048	1024216.03

- The product id 1090 is the highest total sales in this dataset.

10. WHICH PRODUCTS SHOULD BE PROMOTED TO INCREASE REVENUE?

```
select
 ï»¿Product_ID,
 product_category,
 sum(quantity_sold*unit_price) as Total_Revenue
 from productsb
 group by ï»¿Product_ID,product_category
 order by total_Revenue desc;
```

ï»¿Product_ID	product_category	Total_Revenue
1090	Clothing	907378.51
1089	Furniture	868923.5399999999
1028	Clothing	649980.43
1092	Furniture	610704.44
1048	Electronics	600868.25
1032	Electronics	586468.4
1060	Clothing	556938.89
1036	Furniture	554477.9
1019	Clothing	524588.6
1030	Clothing	520543.20000000007
1085	Clothing	511312.41
1097	Electronics	496045.31000000006

- The Clothing product is a promoting the highest revenue.

THANK YOU